How You Can Help

We've mentioned the ESP Game and Google Image Labeler earlier in this article; there are two other games in Luis von Ahn's "Human Computing" initiative that you can play to help computers make the Web a better place:

Phetch (www.peekaboom.org/phetch)

Phetch is like an online treasure hunt, only with images. You find yourself in a group with other users, one of whom is a *describer*; the rest are *seekers*. The describer tells the seekers what kind of image he or she wants to see—"a monkey eating a banana," for instance—and the seekers then use their favourite search engines to hunt down the image. The first seeker to get the right image becomes the next describer.

How it helps: When you play Phetch, you're helping the visually impaired use the web better. When each round is over, the describer's demand is associated with the image that won—much like a caption. So when a visually impaired person is using a text-to-speech converter to read the site that's hosting the image, the caption can be read out to them—"a monkey eating a banana", instead of "monkey.jpg".

Peekaboom (www.peekaboom.org)

Like the others, this game pairs you with a random partner online, and you take your turns as *Peek* and *Boom*. As Boom, you're shown an image and a keyword that represents an object in the image (a family photo, for example, with "boy" as the keyword). You then reveal that part of the image to your partner, Peek, and he or she will start guessing what the keyword is. When Peek gets it right, the round is over, and you switch roles for the next round.

How it helps: Peekaboom is supposed to be used to train computer vision algorithms. Right now, computers recognise objects using basic geometry, but this isn't working too well. If you show a computer lots of images of an object, taken at different sizes and angles, it's possible that they may recognise the object better. This is exactly what you're doing when you isolate objects for your partner to guess. In addition, you're also helping add more labels to the image, hence indirectly contributing to better image searches.

Verbosity (www.peekaboom.org/verbosity)

Think of Verbosity as the online version of word- and phrase-association games like Taboo. You and your partner alternately play the roles of *guesser* and *narrator*. The narrator is given a word ("water", for example), which he or she has to explain to the guesser using clues ("quenches thirst," for example). When the guesser gets it right, roles are reversed.

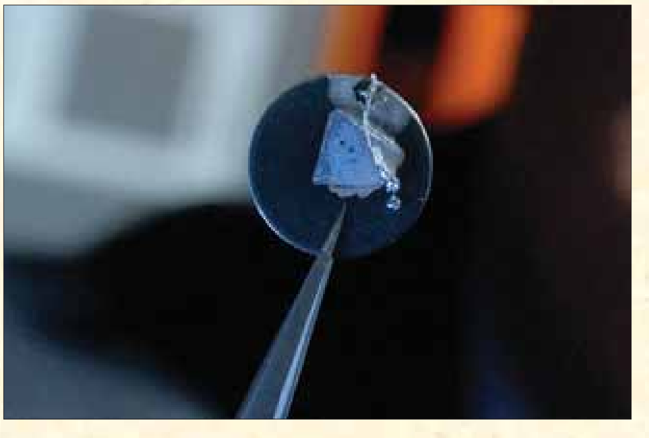
How it helps: Common sense. We have it, computers don't. By looking at the clues that you give your partner, a computer gets a database of "common sense facts" that we take for granted—like "water quenches thirst." This should cut dow on the time it iakes learning machines to learn the most basic things in the world.

I Vant Yer Blud!

When your body needs energy, it uses the glucose in your blood, which, in turn, comes from the food you eat. Back in 2003, Panasonic's Nanotechnology Research Laboratory developed a prototype device that synthesised the glucose in human blood to generate power, the same way your body does. If it were to synthesise all the glucose, it could theoretically generate 100 watts of power, but that would mean you'd have to be eating constantly to compensate for the loss to your body. To keep the balance, the device won't be allowed to consume too much sugar, thus putting a cap on how much power it'll be able to generate, so don't expect this to power your cell phones and PMPs. So you can't go about killing yourself for louder music just yet.

At the other end of the world, Zhong Lin Wang of the Georgia Institute of technology is working on a "nanogenerator" that will use the flow of human blood to generate electricity. It's based on the principle of *piezoelectricity*—the property of some materials that can convert mechanical energy to electrical, and vice versa. The material in this case is a nanowire gird made of Zinc Oxide; the prototype is stimulated by ultrasonic waves, but according to the professor, they can also be made to react to blood flow and muscle contractions.

At this stage, the device only generates nanoamperes of current—enough only to power, well, nano-devices. Still, there is hope—it's been proved that by increasing the size of the nanowire array, the generator can pump out more current—enough to even power personal electronics.



The nanowire array might end the age of recharging batteries for personal gadgets

Think about it—a cell phone that will die only when you do.

All About You

Perhaps some day Subconscious Computing will advance to a point where data can be fed to your brain and processed without your knowing it. If you don't want to wait that long, you can start contributing to making machines a wee bit smarter right now! Read the box below, go online and get started! ⊠

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